Social Network Application With GUI

Project Description

We need to develop a social network application.

Each **User** in the application has a unique ID and name. He may have other users in his friend list.. He has a list of pages which he has liked.

Each **Page** has a unique ID, an owner (user), title, list of posts it has shared (page's timeline) and total number of people who have liked it.

Each **Post** has a unique ID, description/text, no of likes, list of people who have liked it and shared date for example; a post is shared at 25th October 2017. A post can have 10 comments at max. You may optionally allow for an image to be posted as well.

Activity may be a part of the Post, having a type and a value against this type. See table below for details:

Type		Value
1	feeling	Happy/sad/excited etc.
2	thinking about	life/future/meaning of life etc.
3	Making	Money/art/memories etc.
4	celebrating	A birthday/Halloween/success etc.

A **Memory** is a Post having pointer to original post.

A **comment** is posted by a specific user or a page on a single post. All the Users, Pages and Posts in the application are **Objects** having IDs.

Required Functionality:

1- **Set Current User**
   a. Ask user about his ID and set him as current user of application. Now everything that we see in the system will be view of this user.
2- **View Home of Current User**
   a. Display posts of all friends shared in last 24 hours.
   b. Display posts of all his liked pages shared in last 24 hours.
3- **Like a post**
   a. Ask user about ID of post and like it. A post can be liked by 10 users at max.

4- **View the list of People who liked a post**
    a. Take Post ID from user and display list of people who have liked it
5- **Comment on a Post**
    a. Take post ID and comment from the user and add this comment in the post
6- **View a Post**
    a. Take post ID and Display the post with its Comments
7- **Share a Memory**
    a. Take post ID, text from user and share it
8- **View User's Profile**
    a. Display User Name
    b. Display user's posts in his profile
9- **View Friend List**
    a. Take ID of a user and display his friends list
10- **View Page**
    a. Take Page ID and display all its posts

Execution Flow Control: Your application must have a defined entry point. You may create a manager class or an App class to serve as the master or driver class of the application. Make a function SocialNetworkApp::Run() in main. Crucially, the application's functionality should not rely on code directly implemented within the main() function. Delegate all responsibilities to the appropriate classes (separation of concerns).

Important Instructions

- You are not allowed to use global variables and functions
- Use best OOP practices studied so far
- Memory deallocation carries significant weightage
- All the arrays should be dynamically allocated.
- Wherever you need an array of objects, keep it as a double pointer. If a page has no posts or if a user has no friends and no liked pages then these double pointers should be null.

Output highlighted in Blue. U7 is just a sample user, your program should properly run on any user.

Command: Set current System Date 15 11 2017
System Date: 15/11/2017

Command: Set current user "u7"
Abdullah Nadeem successfully set as Current User

Command: View Friend List

Abdullah Nadeem – Friend List

u3 - Waheed Khan
u11 – Ahsan Ali
u12 – Ali Ahmed

Comand: View Liked Pages

Abdullah Nadeem – Liked Pages

P1 – Food Fusion
P2 – Al Jazeera English
P5 – Rumi Quotes
P9 – BBC News

Command: View Home

Abdullah Nadeem – Home Page

//This is displaying posts of 15th and 14th November (15th Nov is currentDate)

--- Waheed Khan is feeling excited
"OOP Project Started Yayy..."
Amna Akram wrote: "Best of Luck"
Abdullah Nadeem wrote: "Interesting"

--- Ahsan Ali shared "Waiting for Weekend"
Saad Ali wrote: "Any plans for Get Together?"
Abdullah Nadeem wrote: "Yes BBQ"
Nimra Batool wrote: "Lunch Buffet please"

--- Ali Ahmed shared "Waiting for Result"

--- Food Fusion is making Kashmiri Chai (Pink Tea).

"Try this at home..."
Food Fusion wrote: "Followers please try this recipe and give your feedback"
Amna Akram wrote: "Yummy"
Ahsan Ali wrote: "One pink tea for me"
--- Al Jazeera English shared "Oil drilling in Alaska's indigenous communities is on the table again."

--- Rumi Quotes shared "Be free of possessing things - Rumi"
Ali Ahmed wrote: "Rumi is great"

//Notice the format of printing Activity and Simple Posts

Command: View Profile

Abdullah Nadeem –Time Line

--- Abdullah Nadeem is Celebrating a birthday
"Happy Birthday to my dear Mom" … (14/11/2017)
Waheed Khan wrote: "Happy Birthday Aunty"

--- Abdullah Nadeem is Celebrating a birthday
"Happy Birthday to my dear Father" … (14/10/2017)
Ahsan Ali wrote: "Happy Birthday Uncle"

--- Abdullah Nadeem is Celebrating Success
"Got D in toughest subject… Thanks God I am Pass finally…" … (15/11/2015)
Ahsan Ali wrote: "Celebrations???"

Command: ViewLikedList(post5)

Post Liked By:
U2 – Amna Akram
P1 – Food Fusion
U3 - Waheed Khan
U4 - Burhan Tahir

Command: LikePost(post5)
Command: ViewLikedList(post5)

Post Liked By:
U2 – Amna Akram
P1 – Food Fusion
U3 - Waheed Khan
U4 - Burhan Tahir
u7 – Abdullah Nadeem

Command: PostComment(post4, Good Luck for your Result)
Command: ViewPost(post4)

--- Ali Ahmed shared “Waiting for Result”
Abdullah Nadeem wrote: “Good Luck for your Result”

Command: PostComment(post8, Thanks for the wishes)
Command: ViewPost(post8)

--- Abdullah Nadeem is Celebrating a birthday
“Happy Birthday to my dear Mom” … (14/11/2017)
Waheed Khan wrote: “Happy Birthday Aunty”
Abdullah Nadeem wrote: “Thanks for the wishes”

Command: SeeYourMemories() //Imp: Sharing a memory multiple times??

We hope you enjoy looking back and sharing your memories on Facebook, from the most recent to those long ago.

On this Day
2 Years Ago
--- Abdullah Nadeem is Celebrating Success
“Got D in toughest subject… Thanks God I am Pass finally…” … (15/11/2015)
Ahsan Ali wrote: “Celebrations???”

Command: ShareMemory(post10, “Never thought I will be specialist in this field…”)
Command: View TimeLine

Abdullah Nadeem –Time Line

~~~ Abdullah Nadeem shared a memory ~~~ …(15/11/2017)
“Never thought I will be specialist in this field…”
(2 Years Ago)
--- Abdullah Nadeem is Celebrating Success
“Got D in toughest subject… Thanks God I am Pass finally…” … (15/11/2015)

--- Abdullah Nadeem is Celebrating a birthday
“Happy Birthday to my dear Mom” … (14/11/2017)
Waheed Khan wrote: “Happy Birthday Aunty”

--- Abdullah Nadeem is Celebrating a birthday
“Happy Birthday to my dear Father” … (14/10/2017)
Ahsan Ali wrote: “Happy Birthday Uncle”

--- Abdullah Nadeem is Celebrating Success
“Got D in toughest subject… Thanks God I am Pass finally…” … (15/11/2015)
~~~

Ahsan Ali wrote: “Celebrations???”

Command: ViewPage(p1)

Food Fusion
--- Food Fusion is making Kashmiri Chai (Pink Tea). … (14/11/2017)
“Try this at home...”
Food Fusion wrote: “Followrs please try this recipe and give your feedback”
Amna Akram wrote: “Yummy”
Ahsan Ali wrote: “One pink tea for me”

--- Food Fusion is making Halwa Puri. … (10/11/2017)
“Sunday Special Breakfast for our followers...”

Command: Set current user “u11”
Ahsan Ali successfully set as Current User
Command: ViewHome
Command: ViewTimeline